

Hypothesis Testing - Complete Study Notes

1. What is Hypothesis Testing?

A statistical method used to decide whether evidence from a sample supports or rejects a claim about a population.

2. Hypotheses

H0 (Null Hypothesis): Assumes no effect, no difference, or no change (default assumption).

H1 (Alternative Hypothesis): Assumes an effect, difference, or change exists.

3. Example

H0: Average marks ≤ 26

H1: Average marks > 26

4. One-tailed vs Two-tailed Tests

One-tailed: Tests only one direction (increase or decrease).

Two-tailed: Tests for any difference.

5. p-value

If $p < 0.05 \rightarrow$ Reject H0.

If $p \geq 0.05 \rightarrow$ Fail to reject H0.

6. Significance Level (Alpha, α)

Usually $\alpha = 0.05$.

7. Type I and Type II Errors

Type I Error = Rejecting a true H0 (False Positive).

Type II Error = Failing to reject a false H0 (False Negative).

8. Confidence Interval (CI)

A range of likely values for the population parameter.

9. Z-test

Used when sample size ≥ 30 or population variance is known.

10. t-test

Used when sample size < 30 and population variance is unknown.

11. One-sample vs Two-sample Test

One-sample: Compare a sample with a known value.

Two-sample: Compare two groups.

Acronyms

H0 = Null Hypothesis

H1 = Alternative Hypothesis

CI = Confidence Interval

MLE = Maximum Likelihood Estimation

MAP = Maximum A Posteriori